



MINI PROJECT

SpendWise

Student Expense Tracker

A simple Flask + SQLite web application for smarter student spending.

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Introduction

Why SpendWise was developed

CONTEXT

Small expenses add up.

Students often make frequent purchases for food, transport, education, shopping and entertainment without having a clear view of total monthly spending.

- Difficult to track many small transactions
- Hard to compare spending with a monthly limit
- Limited visibility into category-wise spending

SOLUTION

One clean dashboard.

SpendWise records expenses, manages a monthly budget, calculates remaining balance and visualizes category-wise spending.

TRACK

Expenses

CONTROL

Budget

INSIGHT

Analytics

Problem Statement

The practical problem addressed by the system

- 01 Tracking** Manual tracking of daily expenses can be inconsistent.
- 02 Visibility** Students may not know which categories consume the most money.
- 03 Budget** Spending can exceed the intended monthly limit without timely awareness.

Objectives

What the project is designed to achieve

01

Record expenses

Store amount, category, description and date.

02

Manage budget

Set and update a monthly spending limit.

03

Calculate balance

Automatically calculate total spent and remaining budget.

04

Analyze spending

Display category-wise totals and progress.

05

Improve usability

Provide a responsive, polished dashboard with light/dark themes.

Technology Stack

Technologies used to implement SpendWise

HTML

Frontend structure

Forms, dashboard layout and dynamic templates.

CSS

UI / UX

Responsive layout, 3D cards, animations and themes.

Python

Programming

Application logic and data processing.

Flask

Backend

Routes, requests, templates and application server.

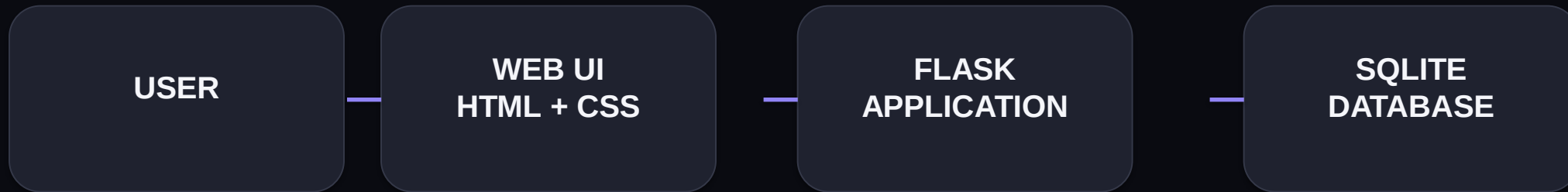
SQLite

Database

Persistent storage for expenses and budget settings.

System Architecture

How the application components interact



User actions → Flask routes → SQLite operations → calculated values → rendered dashboard

Key Features

Core functionality implemented in the current version

ADD EXPENSE

Create a transaction with amount, category and description.

DELETE

Remove an existing transaction.

BUDGET

Set and update the monthly spending limit.

ANALYTICS

View category-wise totals and percentage breakdown.

WARNINGS

Healthy, caution, critical and exceeded budget states.

THEMES

Switch between premium light and dark themes.

RESPONSIVE

Dashboard adapts to phone and desktop layouts.

PERSISTENCE

Expense and budget data are stored in SQLite.

Database Design

SQLite tables used by the application

EXPENSES

id	INTEGER • PRIMARY KEY
amount	REAL • NOT NULL
category	TEXT • NOT NULL
description	TEXT • NOT NULL
date	TEXT • NOT NULL

SETTINGS

id	
INTEGER • PRIMARY KEY	
monthly_budget	
REAL • NOT NULL	

Application Workflow

Example: adding a new expense



Demonstration

Suggested live demo sequence

01

Open dashboard

Show light theme and main summary cards.

02

Add expense

Enter ₹120 • Food • College Lunch.

03

Show analytics

Point out updated total and category breakdown.

04

Change budget

Lower the budget to trigger a warning state.

05

Switch theme

Demonstrate light ↔ dark mode.

06

Delete transaction

Remove the test expense and show the dashboard update.

Future Scope

Possible enhancements beyond the current mini-project scope

- Edit existing expense records
- Search and filter transactions
- Monthly / yearly reports and charts
- User authentication and personal accounts
- Cloud database and deployment
- Export expenses as CSV or PDF

These are optional extensions; the current version already satisfies the core mini-project objectives.

Conclusion

What SpendWise delivers

RESULT

A complete, focused web application.

SpendWise provides a simple way for students to record expenses, manage a monthly budget, understand spending patterns and receive timely budget warnings.

5 Core technologies

8+ Main features

1 SQLite database

• Thank You •